

Project 1

Median Household Statistics

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Math-5565

Data Description

The dataset used in this project contains information for United States ZIP Code Tabulation Areas (ZCTAs). The variable selected for analysis is **Married-Couple Families Median Income (Dollars)**.

- Each observation represents the median income of married-couple families within a specific ZIP code area.
- After removing missing and nonpositive values, the final dataset contained **320,962**, a reliable sample for statistical analysis.

For this study, the random variable **X** is defined as:

X = married-couple families median income (in dollars) for a randomly selected ZIP code

This variable allows us to examine how income varies across different regions of the United States.

Summary Statistics

The variable analyzed in this project is **Married-Couple Families Median Income (Dollars)** for a randomly selected ZIP code. The distribution has a :

mean = \$78,029.68

median = \$71,786.00

This suggests that the data are **right-skewed** because the mean is larger than the median.

standard deviation = \$30,703.08 which shows alot of variability among zipcodes

- **First quartile = \$58,750.00**
- **Median = \$71,786.00**
- **Third Quartile = \$89,663.00,**

Therefore, the middle 50% of observed incomes lie between **\$58,750.00** and **\$89,663.00**. The **(IQR)** is **\$30,913.00**.

- **60th percentile = \$77,608**
- **80th percentile = 95,648**

- This means that 60% of ZIP codes have married-couple family median income at or below the 60th percentile value, and 80% are at or below the 80th percentile value. The 60th percentile data is astonishing because that means 60% of Family households have an income of \$77,608 or below.

The **minimum** observed value is **\$2,500.00** and the **maximum** is **\$250,000.00**, giving a **range** of **\$247,500.00**. The data also have **skewness = 1.77** and **kurtosis = 8.34**, which further supports the conclusion that the income distribution is positively skewed with a heavy right tail.

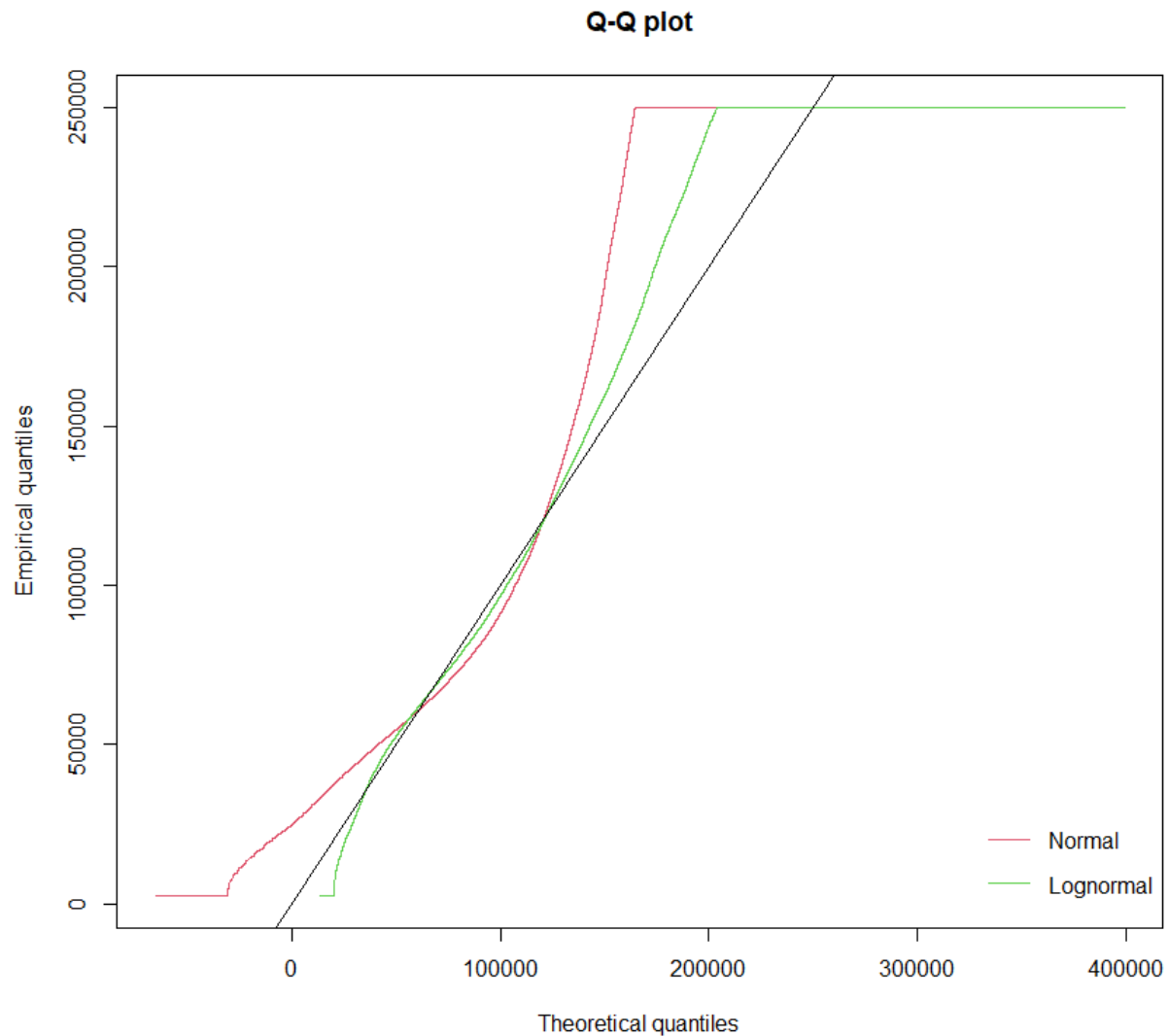
Overall, the exploratory analysis suggests that most ZIP codes fall in a moderate income range, while a smaller number of very high-income ZIP codes pull the distribution to the right.

Q-Q Plot Analysis

A normal quantile–quantile (Q-Q) plot was constructed to assess whether the income data follow a normal distribution. In a Q-Q plot the points that lie close to the reference line indicate that the data are approximately normally distributed.

In this case, the plotted points deviate substantially from the straight line, particularly in the upper tail. Lower quantiles fall below the line, while higher quantiles lie far above it. This pattern indicates strong positive skewness and the presence of unusually large values relative to what would be expected under a normal distribution.

Therefore, the married-couple families' median income data do not follow a normal distribution. The curvature of the plot suggests a right-skewed distribution with a heavy upper tail, which is typical for income data.



Distribution Fitting Using Maximum Likelihood Estimation

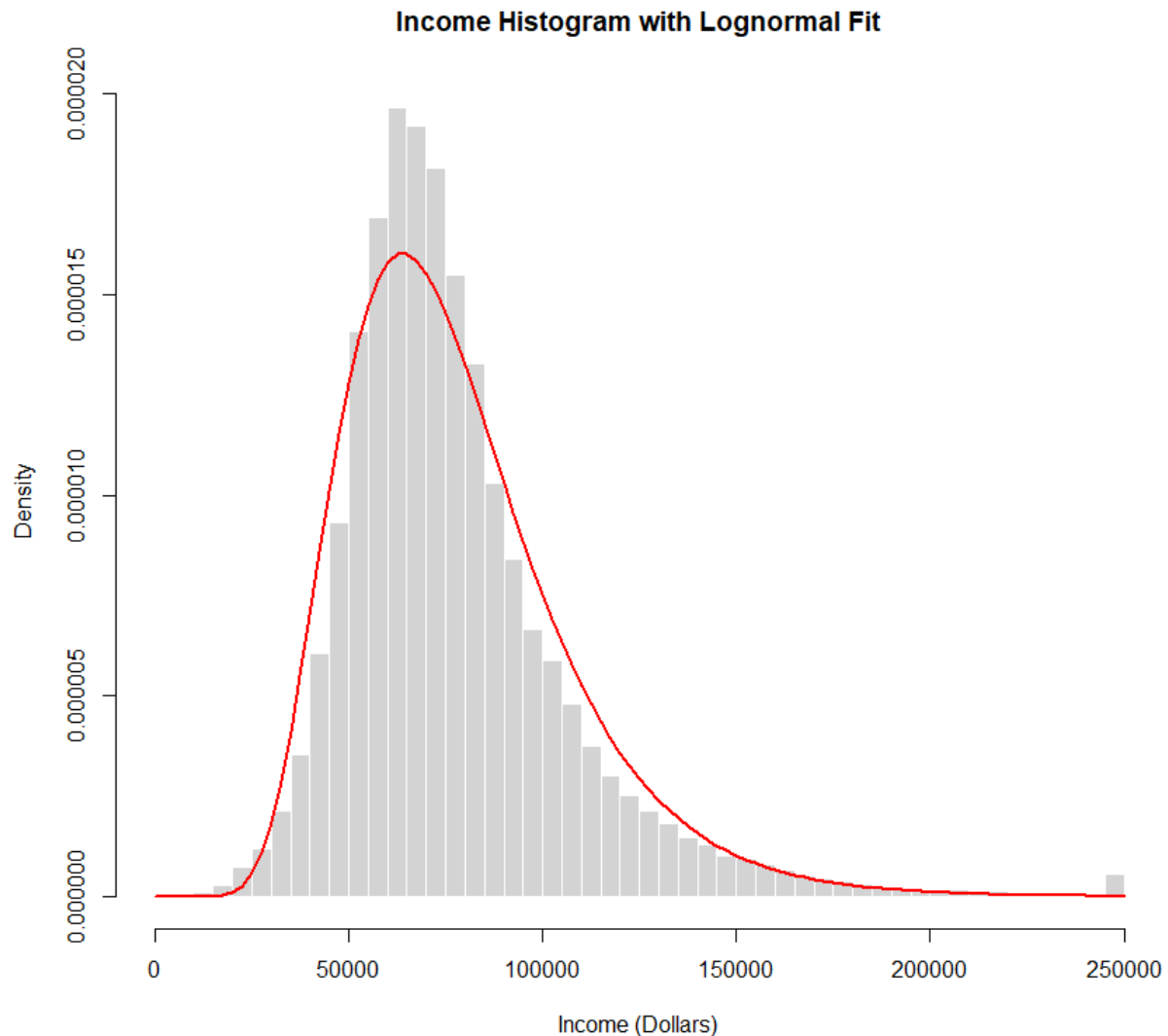
To model the distribution of the income data, maximum likelihood estimation (MLE) was used to fit both a normal distribution and a lognormal distribution.

A histogram of the data shows a unimodal shape with a long right tail, indicating that higher income values occur less frequently but extend far beyond the typical range. When the fitted curves were overlaid on the histogram, the lognormal distribution closely matched the shape of the data, while the normal distribution failed to capture the skewness. As we can see by the Q-Q Plot above, it doesn't do our data justice, while below we can easily interpret the disparities in income.

Additional comparisons using Q-Q plots and goodness-of-fit statistics further confirmed this result. The lognormal model aligned more closely with the observed data, particularly in the upper tail, whereas the normal model showed substantial deviations.

Final Model Conclusion

Better modeled by a lognormal distribution than by a normal distribution.



What I Learned

Through this project, I learned how to work with a real-world dataset using R, from importing data to performing statistical analysis and interpreting results. I gained experience cleaning data, computing summary statistics, and creating visualizations such as histograms and Q-Q

plots. One important lesson was that real data rarely follow idealized assumptions like normality, and graphical tools are essential for understanding the true shape of a distribution.

I also learned how to use maximum likelihood estimation to fit probability distributions and compare models using visual evidence. In this case, the income data were strongly right-skewed, and the lognormal distribution provided a much better fit than the normal distribution.

A major challenge was troubleshooting technical issues, such as correctly importing the dataset, identifying the appropriate variable, handling missing values, and understanding libraries. As well as understanding packages. Finding the right tools to use and understanding what distribution is compatible. Working through these difficulties helped me become more comfortable with R and improved my problem-solving skills when analyzing data.

Overall, this project showed how statistical methods can be applied to real data.

R-Code Used

```
# =====  
# Project 1 - Income Data Analysis  
# Jacob Caro  
# Dataset: us_income_zipcode.csv  
# Variable: Married-Couple Families Median Income (Dollars)  
# =====  
  
library(moments)  
library(fitdistrplus)  
  
# 2. avoid scientific notation  
  
options(scipen = 999)  
  
# --- 3. Import dataset  
  
us_income_zipcode <- read.csv(file.choose(), stringsAsFactors = FALSE)  
  
# -- 4. Check column names  
  
colnames(us_income_zipcode)  
  
# -- 5. Extract variable of interest  
  
x <- us_income_zipcode$Married.Couple.Families.Median.Income..Dollars.
```

```
# -- 6. Clean data
```

```
x <- as.numeric(x)
```

```
x <- x[!is.na(x)]
```

```
x <- x[x > 0]
```

```
# ----- 7. Quick checks
```

```
length(x)
```

```
summary(x)
```

```
# ----- 8. Descriptive statistics
```

```
n_x <- length(x)
```

```
mean_x <- mean(x)
```

```
sd_x <- sd(x)
```

```
min_x <- min(x)
```

```
q1_x <- quantile(x, 0.25)
```

```
median_x <- median(x)
```

```
q3_x <- quantile(x, 0.75)
```

```
max_x <- max(x)
```

```
range_x <- max_x - min_x
```

```
iqr_x <- IQR(x)
```

```
p10_x <- quantile(x, 0.10)
```

```
p60_x <- quantile(x, 0.60)
```

```
p80_x <- quantile(x, 0.80)
```

```
p90_x <- quantile(x, 0.90)
```

```
skew_x <- skewness(x)
```

```
kurt_x <- kurtosis(x)
```

```
# ----- 9. Summary table
```

```
stats_table <- data.frame(
```

```
  Statistic = c(
```

```
    "n", "Mean", "SD", "Min", "Q1", "Median", "Q3", "Max",
```

```
    "Range", "IQR", "P10", "P60", "P80", "P90", "Skewness", "Kurtosis"
```

```
),
```

```
  Value = round(c(
```

```
    n_x, mean_x, sd_x, min_x, q1_x, median_x, q3_x, max_x,
```

```
    range_x, iqr_x, p10_x, p60_x, p80_x, p90_x, skew_x, kurt_x
```

```
), 2)
```

```
)
```

```
stats_table
```

```
# ----- 10. Basic histogram
```

```
hist(  
  x,  
  breaks = 50,  
  main = "Histogram of Married-Couple Families Median Income",  
  xlab = "Income (Dollars)",  
  col = "lightgray",  
  border = "white"  
)
```

```
# ----- 11. Normal Q-Q plot
```

```
qqnorm(x, main = "Normal Q-Q Plot of Income")  
qqline(x, col = "red")
```

```
# ----- 12. Fit distributions using MLE
```

```
fit_norm <- fitdist(x, "norm")  
fit_lnorm <- fitdist(x, "lnorm")
```

```
fit_norm  
fit_lnorm
```

```
# ----- 13. Histogram with lognormal fit
```

```
hist(  
  x,  
  breaks = 50,  
  probability = TRUE,  
  main = "Income Histogram with Lognormal Fit",  
  xlab = "Income (Dollars)",  
  col = "lightgray",  
  border = "white"  
)
```

```
curve(  
  
```

```

dlnorm(
  x,
  meanlog = fit_lnorm$estimate["meanlog"],
  sdlog = fit_lnorm$estimate["sdlog"]
),
add = TRUE,
col = "red",
lwd = 2
)

```

----- 14. Compare normal vs lognormal

```

denscomp(
  list(fit_norm, fit_lnorm),
  legendtext = c("Normal", "Lognormal")
)

```

```

qqcomp(
  list(fit_norm, fit_lnorm),
  legendtext = c("Normal", "Lognormal")
)

```

```

gof_results <- gofstat(list(fit_norm, fit_lnorm))
gof_results

```

----- 15. Print key results

```

cat("Sample size:", n_x, "\n")
cat("Mean:", round(mean_x, 2), "\n")
cat("Median:", round(median_x, 2), "\n")
cat("SD:", round(sd_x, 2), "\n")
cat("Range:", round(range_x, 2), "\n")
cat("P60:", round(p60_x, 2), "\n")
cat("P80:", round(p80_x, 2), "\n")
cat("Skewness:", round(skew_x, 4), "\n")
cat("Kurtosis:", round(kurt_x, 4), "\n")

cat("\nConclusion:\n")

```



```
cat("The income data are strongly right-skewed and are better modeled by a lognormal  
distribution than by a normal distribution.\n")
```